



# Modelling Financial Markets

## Chapter 0: Financial Markets in 2026



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## Acronyms used in this material (1/2)

- ▣ AI: Artificial Intelligence
- ▣ AMM: Automated Market Maker
- ▣ AP: Authorised Participant
- ▣ AR: AutoRegressive (model); in event studies: Abnormal Return
- ▣ ASE: Academia de Studii Economice din București (the Bucharest University of Economic Studies)
- ▣ AUM: Assets Under Management
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BET-TR: BET Total Return (index)
- ▣ BIS: Bank for International Settlements
- ▣ BNR: Banca Națională a României (the National Bank of Romania)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CAGR: Compound Annual Growth Rate
- ▣ CBOE: Chicago Board Options Exchange
- ▣ CCP: Central CounterParty
- ▣ CFTC: Commodity Futures Trading Commission
- ▣ COVID-19: COronaVirus Disease 2019
- ▣ EBA: European Banking Authority
- ▣ EMH: Efficient Market Hypothesis
- ▣ EODHD: EOD Historical Data (market-data provider)
- ▣ ESMA: European Securities and Markets Authority
- ▣ ETF: Exchange-Traded Fund
- ▣ FRED: Federal Reserve Economic Data
- ▣ FX: Foreign eXchange
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GMM: Generalized Method of Moments
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)

## Acronyms used in this material (2/2)

- ▣ HFT: High-Frequency Trading
- ▣ LLM: Large Language Model
- ▣ MDD: Maximum DrawDown
- ▣ MDE: Minimum Detectable Effect
- ▣ MTF: Multilateral Trading Facility
- ▣ NAV: Net Asset Value
- ▣ NYSE: New York Stock Exchange
- ▣ OTC: Over-The-Counter
- ▣ QR: Quick Response (code)
- ▣ SD: Standard Deviation
- ▣ SE: Standard Error
- ▣ SEC: Securities and Exchange Commission
- ▣ SPDR: Standard & Poor's Depositary Receipts
- ▣ SR: Sharpe Ratio
- ▣ US: United States
- ▣ UTC: Coordinated Universal Time (Temps Universel Coordonné)
- ▣ VIX: CBOE Volatility Index

## Chapter Route and Outcomes

### Route of this chapter

- ▣ Part I: the financial system, market structure, measurement and its precision, data conventions, the 2000–2026 landscape
- ▣ Part II: passive investing, digital assets, the Romanian market, a case study, one pass of the AI-assisted research loop
- ▣ Technical detail sits in the Appendix at the end

### After this chapter you can

- ▣ compute returns, CAGR, volatility, drawdowns and correlations on the correct calendar
- ▣ attach a standard error to a mean return, a Sharpe ratio and a correlation change
- ▣ explain how ETFs, stablecoins and market structure shape the data you model

## Course Organisation

### Programme and team

- ▣ Master's programme **Applied Statistics and Data Science**, Faculty of Cybernetics, Statistics and Economic Informatics, ASE
- ▣ 2 hours of lectures + 2 hours of seminars per week, 2026/2027
- ▣ Lecture: Prof. Daniel Traian Pele, [danpele@ase.ro](mailto:danpele@ase.ro)
- ▣ Seminar: Antoaneta Amza, PhD candidate, [antoaneta.amza@csie.ase.ro](mailto:antoaneta.amza@csie.ase.ro)

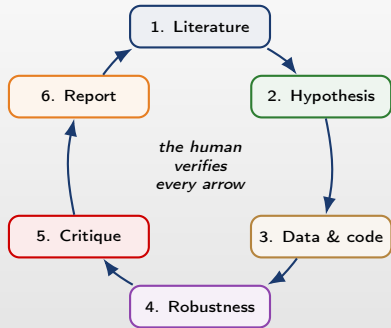
### Grading

- ▣ Team project (GitHub, presentation, oral defence): **70%**
- ▣ Quizzes and seminar activity: **20%**; one quiz per chapter on the course website, 20 questions drawn from a bank, each answer explained
- ▣ Attendance: **10%**; at least **4 lectures** and **4 seminars**, recorded with the code of the day shown in the room (QR code or the Attendance form on the website)
- ▣ No written exam

## The Team Project and Its Outputs

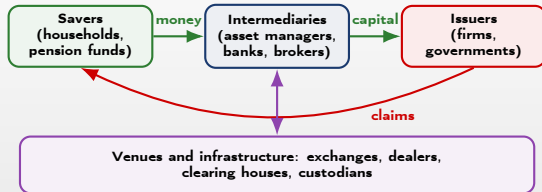
- ▣ Teams of 3–4 students, one research question on real market data; final presentation in Chapter 19
- ▣ One public GitHub repository per team
  - ▶ README (question, data, how to run), weekly commits that document the progress
  - ▶ every figure as a [Quantlet](#): one folder with code, `Metainfo.txt` (name, description, keywords, inputs, outputs) and the chart
- ▣ A result exists only if someone else can reproduce it from the repository
  - ▶ data loaded by code, every reported number produced by a script, random numbers with a fixed seed
- ▣ Tools: Python (pandas, numpy, scipy, statsmodels, scikit-learn, matplotlib), notebooks in Google Colab, the course repository [github.com/danpele/MFM](https://github.com/danpele/MFM)
- ▣ AI use is allowed and declared: every use in [AI\\_USE.md](#), every caught AI error in [AI\\_ERRORS.md](#)

## The AI-Assisted Research Loop



- An AI assistant (an LLM such as Claude, ChatGPT or Gemini) can draft every step: literature map, competing hypotheses, code, robustness plans, a hostile referee report
- The human owns the question, the identifying assumption, the test and every reported number
- Verification checklist
  - ▶ every reference: the DOI (Digital Object Identifier) resolves and the title matches
  - ▶ every number: reproduced from the data by your own code
  - ▶  $H_0$ , test, sample and multiple-testing correction fixed before seeing results
- Typical LLM failures: invented references, plausible but wrong formulas, look-ahead in code, searching until something is “significant”

## What Markets Do and Who Trades



- **Financial market:** a mechanism where buyers and sellers exchange claims on future cash flows (shares, bonds, currencies, contracts, tokens) at prices set by supply and demand
- **Functions:** **capital allocation**, **risk sharing**, **liquidity** (converting assets to cash quickly), **price discovery** (prices aggregate dispersed information), **payments and settlement**
- **Buy side:** retail and institutional investors, asset managers, **passive** (replicate an index) or **active** (try to beat it)
- **Sell side:** brokers execute orders, dealers trade from inventory, **market makers** quote a buy and a sell price

## Asset Classes

| Class          | Examples                                | Cash flow                 | Main risks             |
|----------------|-----------------------------------------|---------------------------|------------------------|
| Equities       | shares, equity ETFs                     | dividends, residual claim | market, firm-specific  |
| Fixed income   | government and corporate bonds          | coupons, principal        | interest rate, credit  |
| Money market   | T-bills (< 1 year), repo (secured loan) | short-term interest       | liquidity              |
| FX             | EUR/USD, EUR/RON                        | interest differential     | currency, policy       |
| Commodities    | gold, oil futures                       | none (roll yield)         | supply, storage        |
| Derivatives    | futures, options, swaps                 | contingent payoffs        | leverage, counterparty |
| Digital assets | Bitcoin, Ether, stablecoins             | none / protocol-specific  | volatility, regulation |

- **ETF** (exchange-traded fund): a fund whose shares trade like a stock; **FX**: the currency market, EUR/RON = lei per euro; **roll yield**: gain or loss from replacing an expiring futures contract
- One statistical toolbox for all of them: returns, volatility, dependence and tails (full list of terms in the Appendix)

## Where Trading Happens

| Venue                                              | How trades are matched                                                            | What is visible                                     | Examples                                |
|----------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------|-----------------------------------------|
| Exchange                                           | central limit order book, auctions                                                | quotes and trades                                   | NYSE, Nasdaq, BVB                       |
| OTC dealer<br>Dark pool                            | bilateral, with a dealer at a price taken from public markets                     | little, after the trade<br>nothing before the trade | bonds, FX, swaps<br>large equity orders |
| Centralised crypto exchange<br>AMM (decentralised) | order book run by a company, 24/7<br>a pool of two tokens, price set by a formula | quotes and trades<br>the pool on the blockchain     | spot Bitcoin, Ether<br>Chapter 16       |

- **OTC**: over the counter, negotiated between two parties without an exchange; **AMM**: automated market maker, a smart contract that trades without an operator
- Crypto trades every day of the year, on hundreds of venues: [Makarov & Schoar \(2020\)](#) find large, persistent cross-exchange price gaps that capital controls and costs keep open

## The Limit Order Book and a Market Order



- **Limit order book:** all waiting buy (**bid**) and sell (**ask**) orders, sorted by price; best bid 99.99, best ask 100.01
- **Bid-ask spread:** best ask minus best bid (0.02), the price of trading at once; **depth:** quantity near the best prices
- A **market buy** of 5 units consumes the book (orange)
  - ▶ 2 units at 100.01 and 3 at 100.02: average  $(2 \times 100.01 + 3 \times 100.02)/5 = 100.016$
  - ▶ the best ask moves to 100.02 and the spread widens to 0.03: **market impact** (Chapter 10)
- Order types: a resting **limit order** provides liquidity; a **market order** consumes it; a **stop order** becomes a market order at a trigger price, and clustered stops can accelerate a fall

## Auctions versus Continuous Trading

### Call auctions

- ▣ Orders are collected, then matched at the one price that maximises traded volume
- ▣ Used at the open and the close of most exchanges
- ▣ Concentrate liquidity: the closing auction sets the daily close used in this course

### Continuous trading

- ▣ Each incoming order is matched at once if possible: best price first, then the earliest order
- ▣ Speed is rewarded: **high-frequency trading (HFT)** programs trade in microseconds
- ▣ Budish, Cramton & Shim (2015) propose frequent batch auctions to end the speed race

## Observation Times and Price Noise

- A “daily price” is a price observed at a specific time
  - ▶ exchange-traded assets: the closing auction (S&P 500: 20:00 or 21:00 UTC)
  - ▶ Bitcoin: no close; the price of day  $t$  is taken at 24:00 UTC, the end of day  $t$
  - ▶ EUR/RON: the BNR reference rate, fixed around 13:00 Bucharest time
- Different observation times split the same news across two days: correlations of assets with different closes are attenuated (Appendix)
- **Bid–ask bounce**: trades alternate between bid and ask, so observed price changes are negatively autocorrelated even when the efficient price is a random walk
  - ▶ the Roll model gives  $\text{Cov}(\Delta p_t, \Delta p_{t-1}) = -s^2/4$  for a spread  $s$  (Roll (1984); derivation in the Appendix)
- Spreads, impact and fragmentation add noise to observed prices, most visible at high frequency (Chapters 9 and 10)

## Returns, Growth and Annualisation

- ▣ **Price index** (S&P 500, BET) ignores dividends; **total-return index** (BET-TR) and **adjusted close** (stocks, ETFs) reinvest them
- ▣ Simple return  $R_t = P_t/P_{t-1} - 1$ ; log return  $r_t = \ln(P_t/P_{t-1}) = \ln(1 + R_t)$ , additive over time
- ▣ **CAGR** (compound annual growth rate)  $= (P_{\text{end}}/P_{\text{start}})^{1/Y} - 1$ , with  $Y = \text{calendar days}/365.25$
- ▣ **Annualised volatility**  $\hat{\sigma}_d\sqrt{q}$ :  $\hat{\sigma}_d$  = standard deviation of daily log returns,  $q$  = observed return intervals per year (about 252 for equities, 365 for crypto)
- ▣ Worked example: S&P 500, 2 Jan 2015 – 18 Sep 2026 (4 277 days,  $Y = 11.71$ )
  - ▶ multiple  $\times 3.72$ :  $\text{CAGR} = 3.72^{1/11.71} - 1 = e^{\ln 3.72/11.71} - 1 = 11.9\%$
  - ▶ with dividends reinvested (SPY, the SPDR S&P 500 ETF): 13.7% a year; the 1.8 percentage point (pp) gap is the dividend yield

## Risk, Dependence and Rates: Notation

- **Drawdown**  $DD_t = P_t / \max_{s \leq t} P_s - 1$ ; **maximum drawdown (MDD)**  $= \min_t DD_t$ , the worst loss from a previous peak
- **Correlation**: Pearson correlation of returns, never of prices; **rolling**: on a moving 252-day window
  - ▶ two calendars (stocks vs crypto): join the *prices* on common days first, then compute returns
- **Yield to maturity**: the internal rate of return of a bond's promised cash flows
  - ▶ equals the realised return only if the bond is held to maturity, pays in full and coupons are reinvested at the same yield
- **Spread**  $y^{10y} - y^{2y}$  in percentage points (pp); 1 basis point (bp)  $= 0.01$  pp; **inversion**: spread  $< 0$

## How Precisely Do We Know an Expected Return?

- Daily returns independent and identically distributed (iid), with mean  $\mu_d$  and standard deviation  $\sigma_d$ ;  $T = qY$  observations over  $Y$  years
- Annualised mean  $\hat{\mu} = q\bar{R}$  has standard error

$$\text{SE}(\hat{\mu}) = \frac{q\sigma_d}{\sqrt{T}} = \frac{\sigma_a}{\sqrt{Y}}, \quad \sigma_a = \sigma_d\sqrt{q}$$

- It depends only on the **span**  $Y$ , not on the sampling frequency ([Merton \(1980\)](#))
- S&P 500 with dividends (SPY), 2015–2026:  $\hat{\mu} = 14.4\%$ ,  $\sigma_a = 17.5\%$ ,  $Y = 11.71$ 
  - ▶  $\text{SE} = 17.5\%/\sqrt{11.71} = 5.1 \text{ pp}$ ; 95% confidence interval [4.3%, 24.4%]
  - ▶ halving the SE needs four times the span (47 years), not more frequent data

## $\sqrt{h}$ Scaling and the Variance Ratio

- Stationary returns with variance  $\sigma^2$  and autocorrelations  $\rho_j$ :

$$\text{Var} \left( \sum_{j=1}^h r_{t+j} \right) = h\sigma^2 \left[ 1 + 2 \sum_{j=1}^{h-1} \left( 1 - \frac{j}{h} \right) \rho_j \right]$$

- The  $\sqrt{h}$  rule is exact when returns are **uncorrelated** with **constant variance**; independence is not needed
- Variance ratio**  $\text{VR}(h) = \text{Var}(h\text{-day return})/(h\sigma^2)$ ; for an AR(1),  $\text{VR}(h) \rightarrow (1 + \rho)/(1 - \rho)$
- S&P 500, 2000–2026:  $\hat{\rho}_1 = -0.10$  and  $\widehat{\text{VR}}(20) = 0.75$ 
  - scaling daily volatility by  $\sqrt{20}$  overstates the 20-day (monthly) risk
  - use the whole autocorrelation function or aggregate the returns directly (more examples in the Appendix; tests in Chapter 2)

## The Sharpe Ratio Is an Estimate

- ▣ **Sharpe ratio (SR)**, risk-free rate set to 0:  $SR_d = \mu_d / \sigma_d$  per day, annualised  $SR_a = \sqrt{q} SR_d$
- ▣ Delta method, iid returns following the Normal distribution (Lo (2002)):

$$\sqrt{T} (\widehat{SR}_d - SR_d) \xrightarrow{d} N(0, 1 + SR_d^2/2) \Rightarrow SE(\widehat{SR}_a) = \sqrt{\frac{1 + SR_a^2/(2q)}{Y}} \approx \frac{1}{\sqrt{Y}}$$

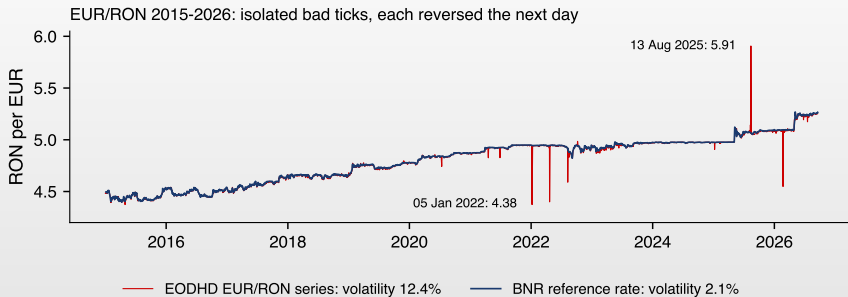
- ▣ SPY, 2015–2026:  $\widehat{SR}_a = 0.82$ ,  $SE = 0.29$ , 95% confidence interval [0.25, 1.39]
- ▣ Like the mean, the SR is known only to about  $\pm 2/\sqrt{Y}$ : 11.7 years cannot separate 0.8 from 0.5
- ▣ Fat tails, dependence and picking the best of many strategies change the SE: Appendix

## Data Used in This Course

| Data                                                 | Content                                         | Format      |
|------------------------------------------------------|-------------------------------------------------|-------------|
| Daily market data, from the course repository        | indices, stocks, ETFs, FX, crypto, yields       | CSV         |
| <a href="#">FRED</a> (Federal Reserve Economic Data) | US rates, exchange rates, macro series          | public CSV  |
| <a href="#">BNR</a>                                  | official RON reference exchange rates           | public XML  |
| <a href="#">DefiLlama</a>                            | stablecoin supply, DeFi (decentralised finance) | public JSON |

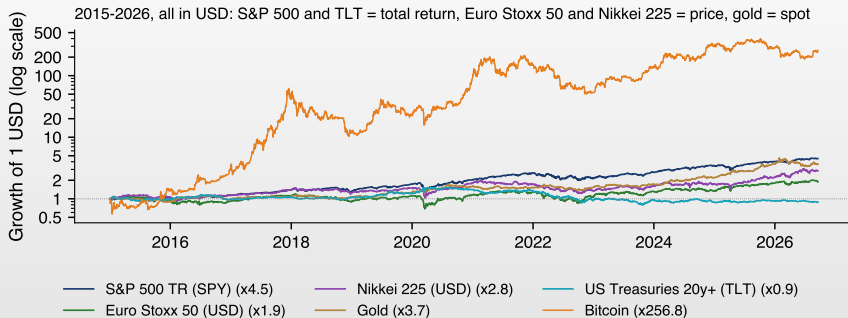
- Stocks and ETFs: adjusted close (dividends, splits); indices, FX, crypto and yields: close
- One series: returns on its own calendar; joint analysis: join the *prices* on common days, then take returns (Friday → Monday for both)
- Annualise with the observed frequency: 365 for crypto, about 260 for spot gold on weekdays, 252 for equities
- **Survivorship**: samples rebuilt from today's survivors, without delisted firms or dead coins, make past returns look too good ([Brown, Goetzmann, Ibbotson & Ross \(1992\)](#))

## Data Pitfalls: a Plotted Bad Tick



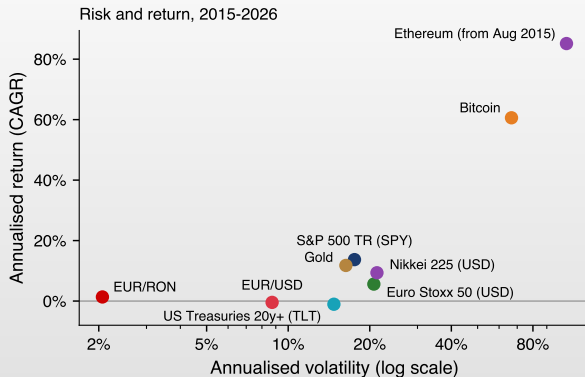
- ❑ **Bad tick:** a wrong price, a spike reversed the next day; EUR/RON series from EODHD (aggregated interbank quotes, not an official rate):  $\pm 15\%$  on 13–14 Aug 2025,  $\pm 12\%$  in early Jan 2022
- ❑ Annualised volatility 2015–2026: 12.4% with these ticks, 2.1% with the BNR reference rate; EUR/RON always from BNR
- ❑ Romania 10-year yield: a 0.00% print on 24 Dec 2013 and stale values on Sundays in late 2022

## Asset Classes since 2015



- Value of 1 USD invested on 2 Jan 2015, log scale; Euro Stoxx 50 and Nikkei 225 converted with daily [FRED](#) exchange rates
- SPY and TLT (iShares 20+ Year Treasury Bond ETF) are total return; Euro Stoxx 50 and Nikkei 225 are price indices, without dividends
- Long Treasuries ended below their 2015 level ( $\times 0.9$ ); Bitcoin's  $\times 257$  is one coin chosen ex post among thousands

## Risk and Return, 2015–2026

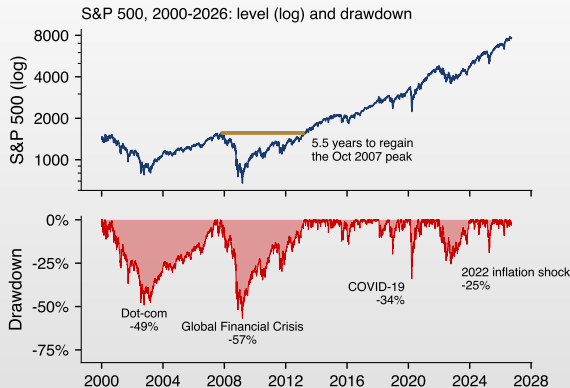


- Vertical axis: CAGR; horizontal: annualised volatility of daily log returns, with the observed  $q$
- All in USD, except EUR/RON (RON per EUR, volatility 2.1% a year, a managed currency)
- S&P 500 with dividends: 13.7% a year at 17.6% volatility; gold 11.8% at 16.3% (weekday quotes)
- Bitcoin 60.6% at 66.7%; Ethereum, from Aug 2015 only, 85.2% at 106%
- Question: is any ranking of these points statistically defensible? (next slide)

## Is Any Ranking in the Risk–Return Chart Significant?

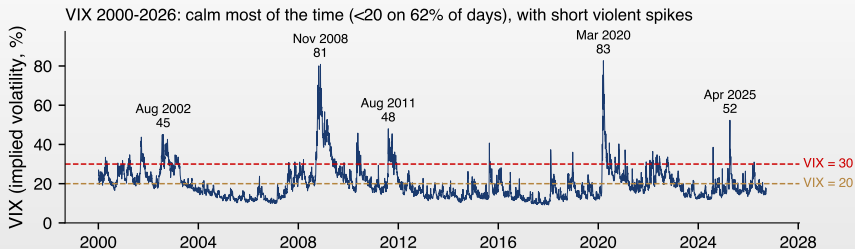
- Standard error of an annualised mean return,  $\sigma_a/\sqrt{Y}$  with  $Y = 11.7$ 
  - ▶ S&P 500 with dividends 5.1 pp, gold 4.8 pp, Euro Stoxx 50 6.0 pp, Nikkei 225 6.2 pp, Bitcoin 19.4 pp
  - ▶ S&P 500 minus gold on common days: +1.9 pp a year, SE 6.8 pp,  $t = 0.28$
- Sharpe ratios: Bitcoin 1.05, S&P 500 0.82, gold 0.77, Nikkei 225 0.53, Euro Stoxx 50 0.37, TLT 0.00, each with SE  $\approx 0.29$
- Overlapping intervals do not test a difference: bootstrap the paired difference on common days (Seminar 0, B1 Extended)
  - ▶ S&P 500 minus gold: +0.06, 95% interval  $[-0.65, +0.81]$
  - ▶ S&P 500 minus Euro Stoxx 50: +0.46,  $[0.09, 0.85]$ , significant alone; none of the 10 pairs of non-crypto assets survives a Bonferroni correction (each test at 0.5%)
- The chart describes the sample; it does not rank expected returns

## Crises and Drawdowns



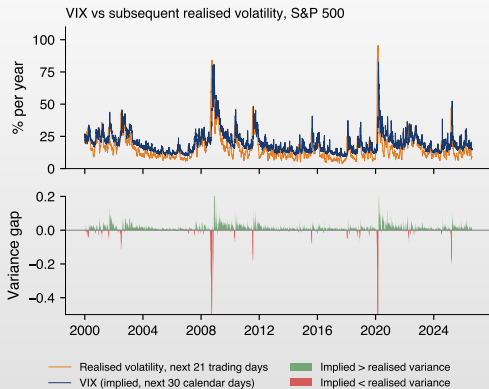
- Lower panel:  $DD_t = P_t / \max_{s \leq t} P_s - 1$  of the S&P 500 (price index)
- Four troughs since 2000: dot-com  $-49\%$  (Oct 2002), global financial crisis  $-57\%$  (Mar 2009), COVID-19  $-34\%$  (Mar 2020, the fastest), inflation shock  $-25\%$  (Oct 2022)
- 1565.15 (9 Oct 2007) to 676.53 (9 Mar 2009); the peak was regained only on 28 Mar 2013
- The MDD deepens with the sample length: compare MDDs only over the same horizon (Appendix)

## Volatility Regimes: the VIX



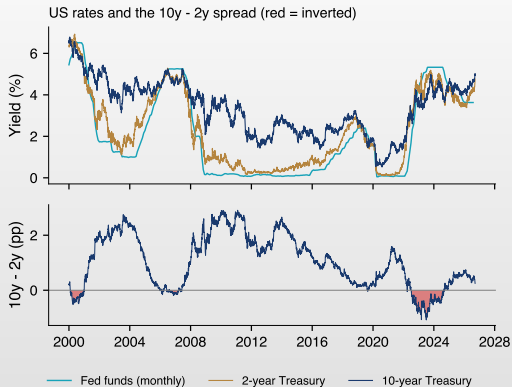
- ▣ **VIX** (CBOE Volatility Index): the S&P 500 volatility over the next 30 days **implied** by option prices, in % per year
- ▣ Since 2000: below 20 on 62% of days, above 30 on 9%; peaks 82.7 (16 Mar 2020), 80.9 (20 Nov 2008), 52.3 (8 Apr 2025)
- ▣ Long calm periods, short intense stress: volatility comes in regimes (Chapters 5 and 9)

## Implied versus Realised Variance



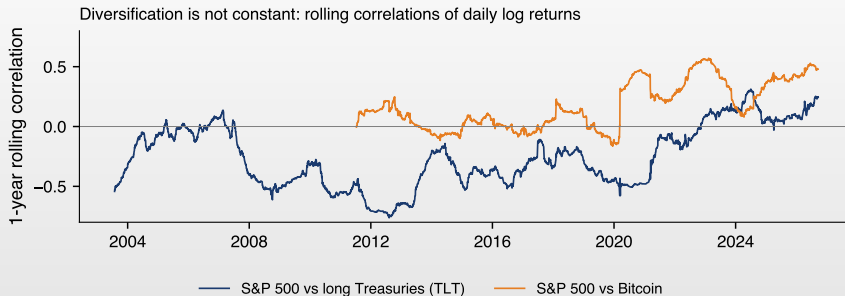
- $(\text{VIX}/100)^2 \approx$  the **risk-neutral** expectation of the annualised S&P 500 variance over the next 30 days
- **Variance risk premium:** risk-neutral minus real-world expected variance, positive on average ([Carr & Wu \(2009\)](#))
- Proxy here: realised volatility over the next 21 trading days,  $\sqrt{(252/21) \sum r^2}$ ; incomplete final windows omitted
- 2000–2026: mean VIX 19.8 vs realised 16.2; the VIX was higher on 83% of days
- Ex post the gap is premium plus forecast error; [Bollerslev, Tauchen & Zhou \(2009\)](#) show that a proxy for the variance risk premium, known at  $t$ , predicts quarterly equity returns

## Interest Rates and the Yield Curve



- **FRED**: 10-year and 2-year US Treasury yields; **fed funds rate**, the overnight rate targeted by the US central bank
- Spread 10y — 2y: briefly negative on 1–4 Apr 2022, then inverted from 6 Jul 2022 to 26 Aug 2024 (537 trading days), minimum  $-1.08$  pp (3 Jul 2023)
- 18 Sep 2026: 10-year 5.01%, 2-year 4.76%, spread  $+0.25$  pp = 25 bp; fed funds 3.63% (Aug 2026)
- A yield is not a realised return: TLT lost value in 2015–2026 as yields rose

## Diversification Benefits Change over Time



- 252-day rolling correlations of daily log returns, prices joined on common days first
- Stocks vs long Treasuries: mean rolling  $-0.43$  (2010–2020),  $+0.08$  (2022–2026); whole-period correlations  $-0.47$  and  $+0.11$ , a different estimator
- Stocks vs Bitcoin: mean rolling  $0.01$  (2016–2019),  $0.38$  (2022–2026); fixed-correlation portfolio models miss such shifts (Chapters 4 and 6)

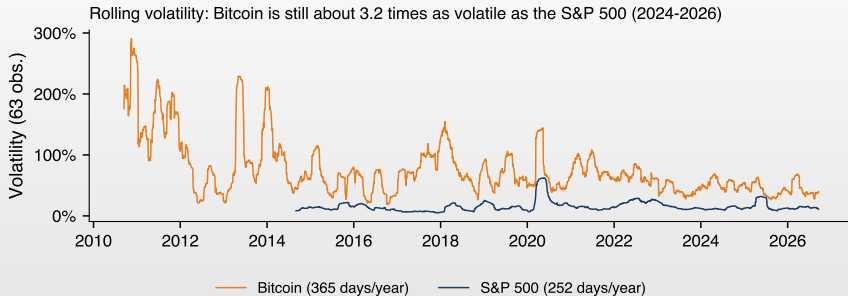
## Has the Stock–Bond Correlation Really Changed?

- ▣ Whole-period correlation S&P 500 – TLT:  $-0.47$  in 2010–2020 ( $n = 2\,769$ ),  $+0.11$  in 2022–2026 ( $n = 1\,182$ ); estimated change  $+0.58$
- ▣ Fisher  $z$  test, iid pairs from a bivariate Normal distribution:  $\operatorname{atanh} \hat{\rho}$  has variance about  $1/(n - 3)$ 
  - ▶  $z = 17.9$ ,  $p < 10^{-10}$ ; iid 95% interval for the change  $[0.52, 0.65]$
- ▣ Volatility clustering makes pairs dependent and fat tails break normality: stationary bootstrap, mean block 20 days
  - ▶ 95% interval  $[0.49, 0.68]$ , 1.54 times wider; the change survives (Seminar 0, B3 Extended)
- ▣ The test establishes a change in association between two periods, not its cause

## Two Traps for Crypto–Equity Correlations

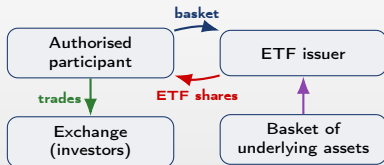
- **Asynchronous closes:** Bitcoin's day ends at 24:00 UTC, the S&P 500's 3–4 hours earlier, so part of the co-movement appears at lags
  - ▶ Bitcoin – S&P 500, 2022–2026: same day 0.42; S&P 500 with Bitcoin one day earlier +0.03, one day later –0.05; sum 0.40
  - ▶ diagnostic: the asynchronous-close bias is small for this pair at the daily frequency
- **Volatility bias:** correlations estimated on turbulent days are larger even when the link is stable ([Forbes & Rigobon \(2002\)](#))
  - ▶ VIX in its top quartile ( $> 21.8$ ): 0.52 vs 0.32 on the other days
  - ▶ adjusted for the 4.64 times higher S&P 500 variance: 0.27, so the “stress correlation” is mostly a volatility effect
- Models, assumptions and correction formulas: Appendix

## Bitcoin Remains Several Times as Volatile as Equities



- Rolling annualised volatility over the last 63 observations: about 3 months for the S&P 500 (annualised with  $\sqrt{252}$ ), about 2 months for Bitcoin (annualised with  $\sqrt{365}$ )
- Bitcoin: 65% on average in 2015–2017, 46% in 2024–2026; S&P 500 in 2024–2026: 14%, so Bitcoin is still about **3.2 times** as volatile
- Volatility changes over time for both: a constant-volatility model is a poor description

## ETFs: Creation and Redemption



- **ETF**: an investment fund whose shares trade on an exchange like a stock; most are **passive** and replicate an index at low cost
- **Net asset value (NAV)**: the fund's assets minus its liabilities, per share, computed once a day
- **Authorised participant (AP)**: a large dealer allowed to deal directly with the issuer
- **Creation**: the AP delivers the basket and receives new ETF shares; **redemption**: the reverse
- This two-way channel keeps the market price close to the NAV; ETFs now hold equities, bonds, commodities and, in the US since January 2024, spot Bitcoin

## Worked Example: Arbitrage Keeps the ETF Price Near NAV

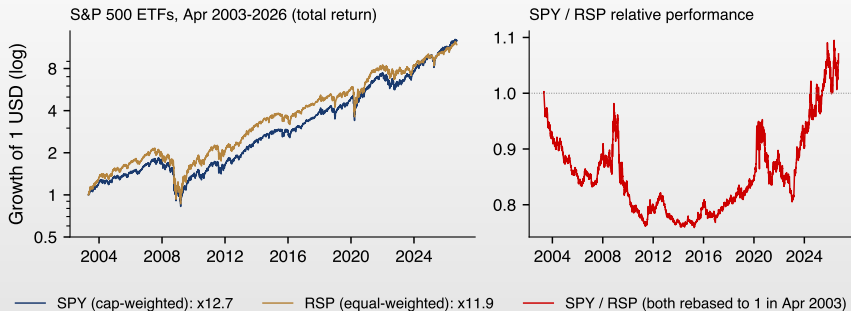
- Illustrative numbers:  $NAV = 100$ , the ETF trades at 100.50; **premium**  $(P - NAV)/NAV = 0.5\%$
- The AP buys the basket for 100, delivers it to the issuer, receives one new ETF share and sells it for 100.50
  - ▶ profit 0.50 per share before costs; the extra ETF shares push the price down until the premium disappears
  - ▶ at a **discount** ( $P < NAV$ ) the AP buys ETF shares and redeems them for the basket
- Premiums persist where the NAV is stale or the basket is illiquid, e.g. foreign stocks or bonds ([Petajisto \(2017\)](#))
- The speed at which premiums decay can be measured with an AR(1) half-life (Appendix)

## Cap-Weighted versus Equal-Weighted Indices

### Definition 1 (Index weights and effective diversification)

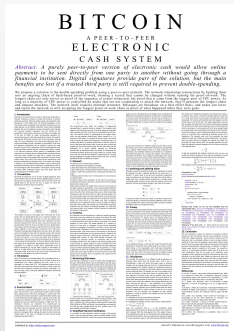
- Capitalisation weights  $w_i = MC_i / \sum_{j=1}^n MC_j$ , market capitalisation  $MC_i = \text{price} \times \text{number of shares}$ ; equal weights  $w_i = 1/n$ , rebalanced periodically
- Effective number of stocks  $N_{\text{eff}} = 1 / \sum_i w_i^2$  (inverse Herfindahl):  $n$  for equal weights, far fewer when a few firms dominate
- SPY** (SPDR S&P 500 ETF): the S&P 500 with capitalisation weights; **RSP** (Invesco S&P 500 Equal Weight ETF): the same stocks, each  $1/500$
- A cap-weighted index is less diversified than its number of stocks suggests; its risk depends on a few firms and themes (Chapters 3, 4 and 17)
- Passive cap-weighted funds hold each firm in proportion to its size, so their investors carry the concentration of the largest firms

## Market Concentration: SPY versus RSP



- ▣ Total return (adjusted close) since RSP's launch in April 2003: SPY  $\times 12.7$ , RSP  $\times 11.9$ ; the ratio ends at 1.07
- ▣ SPY/RSP rose by 31% since January 2023: a handful of very large companies drove the index
- ▣ The ratio measures relative performance; it is not a direct measure of concentration or of passive ownership

## Bitcoin, Ethereum and Tokens



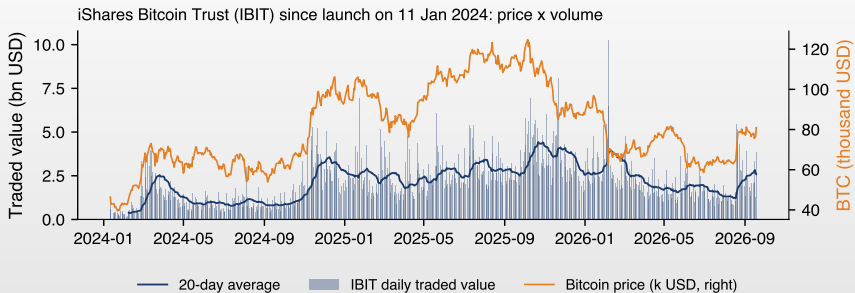
Bitcoin white paper (2008)

Satoshi Nakamoto (2008); CC0; Wikimedia

[Commons](#)

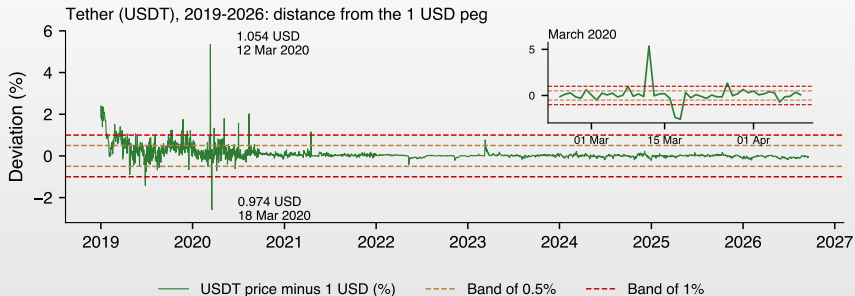
- 31 Oct 2008: *Bitcoin: A Peer-to-Peer Electronic Cash System*, by the pseudonymous Satoshi Nakamoto [Nakamoto \(2008\)](#)
- **Blockchain**: a shared ledger of transactions, copied on many computers and extended in blocks, without a trusted intermediary
- **Bitcoin** (2009): fixed supply rule (at most 21 million coins); **proof of work**: adding a block costs computing power, which makes rewriting history expensive
- **Ethereum**: a blockchain that runs programs (**smart contracts**); its coin is Ether; a **token** is a unit on a blockchain representing a coin, a claim or a right
- [Härdle, Harvey & Reule \(2020\)](#): statistical properties, pricing and regulation; [Pele et al. \(2023\)](#): tail, memory and moment factors set cryptos apart from stocks, FX, commodities, bonds and real estate

## Spot Bitcoin ETFs



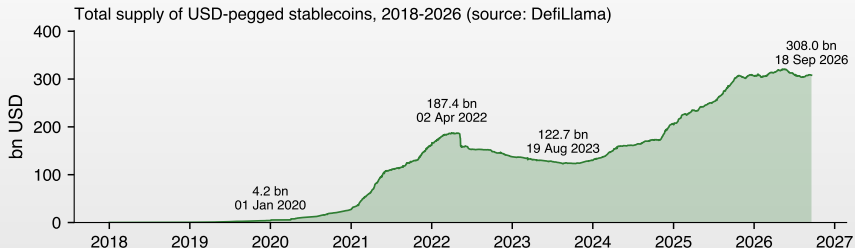
- **Spot** ETF: holds the asset itself, not futures; **ETP** (exchange-traded product): the wider category; spot Bitcoin ETPs approved in the US on 10 Jan 2024 ([SEC \(2024\)](#))
- IBIT (iShares Bitcoin Trust), trading since 11 Jan 2024: daily traded value = price  $\times$  shares traded; median 1.97 bn USD a day, peak 10.3 bn USD (5 Feb 2026)

## Stablecoins and the Peg



- **Stablecoin**: a token designed to keep a fixed value (**peg**), usually 1 USD; **reserve-backed** (cash, T-bills, redeemable at 1 USD) or **algorithmic** (a second token and trading rules); **depeg**: the price leaves the peg
- Tether (USDT), 2019–2026: within  $\pm 0.5\%$  of 1 USD on 93% of days, within  $\pm 1\%$  on 98%; extremes 0.974 and 1.054 USD in March 2020
- The algorithmic stablecoin TerraUSD lost its peg and collapsed in May 2022

## Stablecoin Supply



- **Supply:** the USD value of USD-pegged stablecoins in circulation ([DefiLlama](#)): 4.2 bn (1 Jan 2020), peak 187.4 bn (2 Apr 2022), trough 122.7 bn (19 Aug 2023), 308.0 bn (18 Sep 2026)
- The 2022–2023 fall followed the TerraUSD collapse and the crypto crash; circulation value is distinct from the peg: supply can fall while the price stays at 1 USD

## Tokenisation and Regulation

### Tokenised real-world assets

- ▣ **RWA** (real-world assets): bonds, fund shares, deposits, commodities
  - ▣ A token on a shared ledger is a claim on such an asset: transfer and settlement become programmable and near-instant
  - ▣ Examples: tokenised money-market and Treasury funds
  - ▣ **BIS (2023)**: a “unified ledger”; **BIS (2025)**: stablecoins fail the tests for money (Appendix)
- ▣ Statistical question: do tokenised assets inherit the properties of the underlying, or those of crypto markets?

### Regulation, key dates

- ▣ EU: **MiCA, Regulation (EU) 2023/1114** (Markets in Crypto-Assets)
- ▣ US: spot Bitcoin ETPs approved on 10 Jan 2024 (**SEC (2024)**); payment stablecoins under the **GENIUS Act, Public Law 119-27**
- ▣ Regulatory dates are natural break points and event-study dates

## The Bucharest Stock Exchange Today

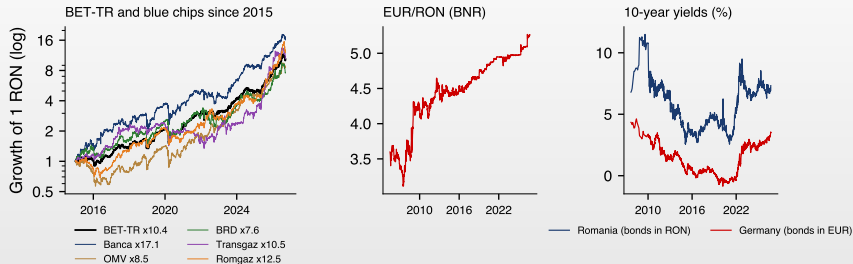
- **BVB** (Bursa de Valori București): opened in 1882, closed in 1948, re-established in 1995 (history in the Appendix)
- **BET** (Bucharest Exchange Trading index): the most traded Romanian companies, launched on 19 Sep 1997 at 1 000 points **BVB (indices)**; **BET-TR**: its total-return version, with reinvested dividends
- Largest companies: banks (Banca Transilvania, BRD), energy (OMV Petrom, Romgaz, Hidroelectrica), utilities (Transgaz)
- Classified as **Secondary Emerging** by **FTSE Russell (2026)** since September 2020
- Milestones: Hidroelectrica, the largest IPO (initial public offering) in BVB history, EUR 1.9 bn, July 2023 **BVB (history)**; **TVBETETF**, an ETF on BVB that replicates BET

## The BET Index, 1997–2026



- Official closing values, log scale; peak 10 813.59 (24 Jul 2007), trough 1 887.14 (25 Feb 2009): −82.5%, regained only on 16 Mar 2021, after 13.6 years
- 19 Dec 2018: −11.2% in one day, the day after the bank-tax announcement; Mar 2020: −31.1% from the February high
- 18 Sep 2026: 32 757.81, ×32.8 since launch, CAGR 12.8% over 29 years (record 36 487.21 on 19 Aug 2026)

## Romanian Equities, the Leu and Government Yields



- Dividend-adjusted growth since Jan 2015: BET-TR  $\times 10.4$ , Banca Transilvania  $\times 17.1$ ; EUR/RON (BNR) from 3.60 (4 Jul 2005, after the redenomination 1 RON = 10 000 ROL) to 5.26
- 10-year yields, 18 Sep 2026: Romania 7.33% (bonds in RON), Germany 3.53% (in EUR): a cross-currency differential of 3.8 pp
- This differential mixes expected leu depreciation, inflation and policy-rate gaps with credit, liquidity and term premia: it is **not** a pure sovereign-credit spread

## Case Study: Question and Elasticity

- ▣ Haddad, Huebner & Loualiche (2025), *American Economic Review* 115(3), 975–1018
- ▣ **Question:** when some investors turn passive, do the others trade more aggressively and undo the effect on prices?
- ▣ **Demand elasticity**  $\mathcal{E}_i$ : the percentage fall in the shares investor  $i$  demands after a 1% price rise; a passive investor has  $\mathcal{E}_i = 0$
- ▣ **Aggregate elasticity**  $\mathcal{E}_{agg}$ : the holdings-weighted average of the  $\mathcal{E}_i$  (eq. 2); a demand shift  $\Delta \bar{d}$  moves the log price by  $\Delta p = \mathcal{E}_{agg}^{-1} \Delta \bar{d}$  (eq. 3)
  - ▶ lower  $\mathcal{E}_{agg}$ : prices more volatile and less informative (Section II.A)
- ▣ Link: stocks with higher ETF ownership are more volatile (Ben-David, Franzoni & Moussawi (2018))

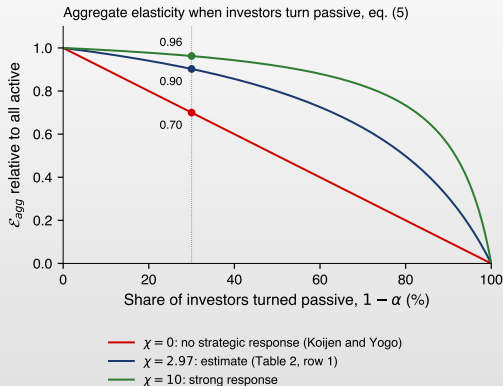
## Case Study: Identification Sketch

- **Strategic response**  $\chi$ :  $\mathcal{E}_i = \bar{\mathcal{E}}_i - \chi \mathcal{E}_{agg}$  (eq. 4), with  $\bar{\mathcal{E}}_i$  the investor's own component
  - ▶  $\chi = 0$ : nobody reacts, the demand system of [Kojien & Yogo \(2019\)](#);  $\chi \rightarrow \infty$ : the others fully offset any change
- **Data**: quarterly holdings of US institutional managers (SEC Form 13F), 2001–2020, about 80% of US market capitalisation
- **Identification**: the same investor meets a different mix of other investors in each stock, which separates  $\bar{\mathcal{E}}_i$  from  $\chi$  (Theorem 1)
  - ▶ instruments: the money that would flow to each stock if the other investors held equal weights in their usual universes
- **Estimation**: GMM (generalised method of moments) with a nested equilibrium; the full demand system, equilibrium conditions and instruments are in the Appendix

## Case Study: Headline Results

- Strategic response  $\chi = 2.97$  (SE 0.47; Table 2, row 1): investors react, far from both 0 and  $\infty$
- Demand for single stocks is inelastic: mean  $\mathcal{E}_{agg} = 0.438$  (Table 3)
- **Pass-through** of a fall in the active share to  $\mathcal{E}_{agg}$  (eq. 28):  $1/(1 + 3 \times 0.68) = 33\%$ 
  - ▶ median active share 81.4% (2001) to 59.5% (2020): a fall of  $\ln(81/59) = 32\%$
  - ▶ times the pass-through: stocks are about 11% less elastic; two-thirds of the direct effect is offset
- Economic implication: inelastic stocks are more volatile and less liquid (Table 5); robustness and reduced-form checks in the Appendix

## Case Study: Strategic Response Damps the Effect of Passive Investing

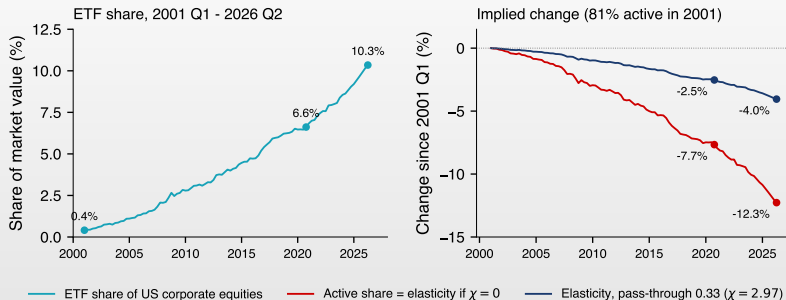


- Eq. (5):  $\mathcal{E}_{agg}$  after a share  $1 - \alpha$  of identical investors turns passive, relative to all active:

$$\alpha + (1 - \alpha) \frac{\chi \alpha}{1 + \chi \alpha}$$

- Worked point, 30% passive:  $\chi = 0$  gives 0.70;  $\chi = 2.97$  gives 0.90, two-thirds of the direct fall offset
- $\chi = 10$  still gives 0.96: only an infinite response keeps  $\mathcal{E}_{agg}$  unchanged
- All curves reach 0 at 100% passive: nobody is left to absorb demand shocks

## Case Study: ETF Share and the Implied Fall in Elasticity



- ETF share of US corporate equities (Financial Accounts of the United States): 0.4% (2001 Q1), 6.6% (2020 Q4), 10.3% (2026 Q2)
- Rule of Section V.A (81% active in 2001, pass-through 0.33): active share  $-7.7\%$  and elasticity  $-2.5\%$  to 2020;  $-12.3\%$  and  $-4.0\%$  to 2026 Q2
- A scenario, not a replication: ETFs include active funds and exclude index mutual funds

## Case Study: Limits and an Open Question

- Limits stated by the authors
  - ▶ one  $\chi$  for all investors; no strategic entry, exit or fund flows (Section I.B)
  - ▶ one 13F form per institution: its active and passive funds cannot be separated (footnote 51)
  - ▶ the elasticity is for single stocks, not for the market as a whole (footnote 37)
  - ▶ those who turn passive are assumed representative of active investors (Section V.A)
- Extension: does the pass-through change if the most elastic investors turn passive first? (Seminar 0, C3)
- Open question: does passive investing change price formation?
  - ▶ [Sammon \(2025\)](#): higher passive ownership makes prices anticipate earnings announcements less
  - ▶ passive ownership is endogenous: identification relies on index reconstitutions and flow shocks

## Hypothesis: Is There a Concentration Premium?

- ▣ A typical assistant proposal: “passive flows favour the largest firms, so cap weights beat equal weights”
- ▣ Discovery loop in one pass
  - ▶ **literature**: ask for peer-reviewed papers with DOIs, then resolve each DOI yourself
  - ▶ **hypothesis**: three competing explanations (passive flows, sector composition, earnings growth of the largest firms), each with one implication the others do not share
  - ▶ **data and code**: SPY and RSP adjusted (total-return) prices, joined on common days; first replicate SPY  $\times 12.7$  and RSP  $\times 11.9$
- ▣ Test:  $d_t = r_{\text{SPY},t} - r_{\text{RSP},t}$  (daily log returns);  $H_0 : \mathbb{E}[d_t] = 0$  against  $H_1 : \mathbb{E}[d_t] \neq 0$
- ▣ Checks fixed in advance: the price type, the calendar join, the windows (full sample and two halves) and the HAC (heteroskedasticity and autocorrelation consistent) Newey–West  $t$ -statistic

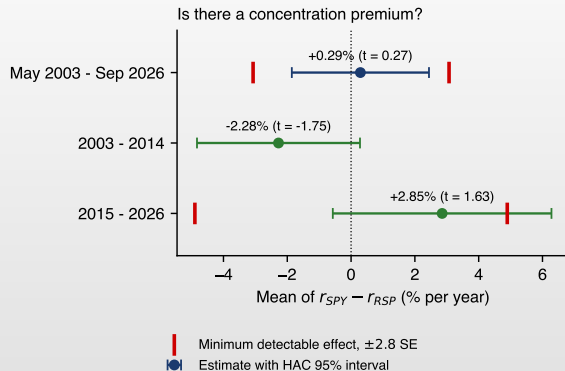
## Estimation and Power

- ▣ May 2003 – September 2026 (5 884 days): mean spread +0.29% a year,  $t = 0.27$
- ▣ Pre-specified halves: 2003–2014 −2.28% a year ( $t = -1.75$ ); 2015–2026 +2.85% ( $t = 1.63$ )
- ▣ **Minimum detectable effect (MDE)**: the smallest true effect found with 80% power at the 5% level

$$\text{MDE} = (z_{0.975} + z_{0.80}) \times \text{SE} \approx 2.8 \times \text{SE}$$

- ▶ full sample: 3.1% a year; 2015–2026: 4.9% a year, larger than the estimate itself
- ▣ “Not significant” is not “no effect”: here the design cannot detect premiums below about 3% a year
- ▣ Causality: SPY − RSP is an outcome, not a measure of passive ownership; a causal claim needs an exogenous shifter

## Mini-Case: Estimate, Interval and MDE



- Dots: annualised mean of  $d_t$ ; bars: HAC 95% intervals; red ticks:  $\pm$ MDE
- Every interval contains 0
- The sign flips between the two halves: the “premium” is a feature of the chosen window
- Report every window tried, not only the one that looks significant

## Your Turn: A Project Seed

- ▣ **Question:** does the SPY – RSP spread survive after controlling for sector returns?
  - ▶ regress  $d_t$  on the sector ETFs (XLK, XLC, XLF, ...) from the course market data; test  $H_0$ : intercept = 0 with HAC standard errors
- ▣ **Replicate first:** SPY  $\times 12.7$ , RSP  $\times 11.9$  and the mini-case  $t$ -statistics
- ▣ **Pre-register:**  $H_0$ ,  $H_1$ , test, sample and corrections, committed before the results
- ▣ **Report:** estimate, standard error, MDE and every specification tried; the code as a Quantlet
- ▣ **Log:** prompts that shaped decisions in AI\_USE.md, every caught error in AI\_ERRORS.md; reusable prompts for each step are in the Appendix

## Key Takeaways

- ▣ Markets in 2026 combine exchanges, OTC dealers, ETFs, algorithms and blockchains; the observation time of a “daily price” depends on the venue
- ▣ 2000–2026 repeats the same patterns: deep drawdowns with slow recoveries, short volatility spikes, correlations that change with the macro regime
- ▣ Every number is an estimate
  - ▶ standard error of an annualised mean return:  $\sigma_a/\sqrt{Y}$ ; of an annualised Sharpe ratio: about  $1/\sqrt{Y}$
  - ▶  $\sqrt{h}$  scaling, correlations and drawdowns rest on assumptions that the data often violate
- ▣ Cap-weighted passive investing carries market concentration; crypto moved closer to mainstream finance through ETFs and stablecoins
- ▣ Data choices (field, calendar, source) are modelling decisions

## Check Yourself, and Next: Chapter 1

### Retrieval questions

- Why does the SE of a mean return not fall when you use daily instead of monthly data?
- Why must prices be joined on common days before computing Bitcoin – S&P 500 returns?
- Why is the SPY/RSP ratio not a measure of passive ownership?

### Chapter 1: stylised facts

- How heavy are the tails of daily returns, and is the kurtosis even finite?
- Does volatility have long memory?
- At which horizon do returns approach the Normal distribution?
- Do the facts hold for the S&P 500, Bitcoin, BVB stocks and EUR/RON?

## Appendix: Course Map, 20 Chapters

- ▣ 0 Financial Markets in 2026
- ▣ 1 Stylised Facts and Returns
- ▣ 2 Market Efficiency: From EMH to Adaptive Markets
- ▣ 3 Factor Models and Asset Pricing
- ▣ 4 Portfolio Optimisation
- ▣ 5 Conditional Volatility: GARCH Models
- ▣ 6 Multivariate Volatility and Dependence
- ▣ 7 Value-at-Risk and Expected Shortfall
- ▣ 8 Backtesting and Evaluating Risk Forecasts
- ▣ 9 Realised Volatility
- ▣ 10 Market Microstructure
- ▣ 11 Continuous-Time Models
- ▣ 12 Options and the Volatility Surface
- ▣ 13 Machine Learning in Finance
- ▣ 14 Deep Learning and Time-Series Foundation Models
- ▣ 15 LLMs and Sentiment Analysis
- ▣ 16 Digital Assets and DeFi
- ▣ 17 Bubbles and Crashes
- ▣ 18 Systemic Risk and Networks
- ▣ 19 Review and Project Presentations

## Appendix: Core Reading and Further Study

- ▣ Franke, Härdle & Hafner (2019), *Statistics of Financial Markets*, 5th ed.: the main statistical reference
- ▣ Campbell, Lo & MacKinlay (1997), *The Econometrics of Financial Markets*; Brandimarte (2018), *An Introduction to Financial Markets: A Quantitative Approach*
- ▣ Härdle & Simar (2019), *Applied Multivariate Statistical Analysis*; Härdle, Chen & Overbeck (2017) (Eds.), *Applied Quantitative Finance*, 3rd ed.
- ▣ Härdle, Harvey & Reule (2020), "Understanding cryptocurrencies", *Journal of Financial Econometrics*
- ▣ Advanced courses on [quantinar.com](https://quantinar.com)
  - ▶  Statistics of Financial Markets;  Digital Transformation on Finance
  - ▶  Cryptos, NFTs, Digital Assets;  Introduction to Blockchain and Cryptocurrencies

## Appendix: Asset-Class Terms

- ▣ **T-bills:** Treasury bills, government debt with maturity below one year; **repo:** repurchase agreement, a short-term loan backed by securities
- ▣ **Futures:** a contract to buy or sell later at a price fixed today; **roll yield:** the gain or loss from replacing an expiring contract with the next one
- ▣ **Options:** the right, not the obligation, to buy or sell at a fixed price; **swaps:** exchanges of cash-flow streams
- ▣ **Idiosyncratic risk:** specific to one company; **counterparty risk:** the other side of a contract may not pay
- ▣ **Multilateral trading facility (MTF):** a trading venue run outside a regulated exchange; **central counterparty (CCP):** becomes the buyer to every seller and the seller to every buyer; **custodian:** keeps the securities

## Appendix: From the Trading Floor to Screens



NYSE trading floor, 2011

Photo: Carol M. Highsmith (2011);  
public domain; Wikimedia Commons



Electronic displays, Chicago Board of Trade, 1992

Photo: Lars Plougmann (1992); CC BY-SA 2.0;  
Wikimedia Commons

- **Open outcry:** traders met on the floor and matched orders by voice and hand signals
- Electronic limit order books gradually replaced most floor trading
- Today most volume is matched by computers (market microstructure, Chapter 10)

## Appendix: A Historic Crash, Black Tuesday, 1929



Crowd outside the NYSE, 29 October 1929

Photo: U.S. government (1929); public domain;  
[Wikimedia Commons](#)

- 29 October 1929 (**Black Tuesday**): crowds gather outside the New York Stock Exchange during the crash
- The crash of October 1929 deepened a US recession that had begun in August 1929; the initial contraction of the Great Depression lasted until 1933 ([NBER business-cycle dates](#))
- Long, deep drawdowns are not new; bubbles and crashes are modelled in Chapter 17

## Appendix: The Roll Model of Bid-Ask Bounce

- Model:  $p_t = m_t + (s/2) q_t$ , with  $m_t$  the efficient price (a random walk),  $s$  a constant spread in price units,  $q_t = \pm 1$  independent, equally likely trade signs, unrelated to  $\Delta m_t$
- Then  $\Delta p_t = \Delta m_t + (s/2)(q_t - q_{t-1})$  and only the shared  $q_{t-1}$  links consecutive changes:

$$\text{Cov}(\Delta p_t, \Delta p_{t-1}) = -\frac{s^2}{4} \text{Var}(q_{t-1}) = -\frac{s^2}{4}, \quad \hat{s} = 2\sqrt{-\hat{\gamma}_1}$$

- If  $\hat{\gamma}_1 > 0$  the estimator is undefined ( $\hat{\gamma}_1 = 0$  gives  $\hat{s} = 0$ ); other dependence also moves  $\hat{\gamma}_1$ , so not every negative autocovariance is bounce ([Roll \(1984\)](#); Chapter 10)

## Appendix: Variance Ratios and Variance Estimates, More Examples

- EUR/RON, BNR reference rate, 2005–2026:  $\hat{\rho}_1 = 0.17$ 
  - ▶ a hypothetical AR(1) would give the factor  $\sqrt{1.17/0.83} = 1.19$  (the 2005–2026 volatility annualised with  $\sqrt{252}$ , 4.4%, would become 5.3%)
  - ▶ but  $\hat{\rho}_2, \hat{\rho}_3, \hat{\rho}_4 < 0$  offset lag 1: monthly returns give 4.4% (Seminar 0, A6)
- EUR/RON series from EODHD: bad ticks give  $\hat{\rho}_1 = -0.57$  and inflate  $\sqrt{252}$  volatility; they are errors to remove, not an AR(1)
- Variance behaves differently from the mean:  $\text{Var}(\hat{\sigma}^2) \approx \sigma^4(\kappa - 1)/T$  shrinks with  $T$ , slowly when the kurtosis  $\kappa$  is large (SPY daily:  $\kappa = 17.1$ , Chapter 1)

## Appendix: The Sharpe Ratio under Fat Tails, Dependence and Selection

- Non-Normal iid returns with skewness  $\gamma_3$  and kurtosis  $\gamma_4$ : the asymptotic variance of  $\sqrt{T}(\widehat{\text{SR}}_d - \text{SR}_d)$  is

$$V = 1 - \text{SR}_d \gamma_3 + \text{SR}_d^2 (\gamma_4 - 1)/4, \quad \text{SE}(\widehat{\text{SR}}_d) = \sqrt{qV/T}$$

- ▶ Opdyke (2007); SPY: SE 0.30 instead of 0.29, because the daily SR is small
- Dependent returns: HAC or studentised bootstrap inference (Ledoit & Wolf (2008))
- Selection: picking the best of many assets or strategies ex post inflates the SR; the deflated SR corrects for it (Bailey & López de Prado (2014), Chapter 13)

## Appendix: Asynchronous Closes

- Bitcoin's price for day  $t$  is taken at 24:00 UTC; the S&P 500 closes at 20:00 or 21:00 UTC: news arriving between the two times enters the two series on different days
- With a common factor, the same-day correlation is attenuated and the rest appears in lead/lag correlations
  - ▶ if only the common factor is split across days and daily variances are unchanged, the lead, same-day and lag correlations sum to the true one (Scholes & Williams (1977); Lo & MacKinlay (1990); Seminar 0, A9)
  - ▶ otherwise use longer returns (Burns, Engle & Mezrich (1998))
- Bitcoin – S&P 500, 2022–2026, common days:  $0.42 + 0.03 - 0.05 = 0.40$ ; the sum is a diagnostic, not a proof

## Appendix: Correlation in Turbulent Periods, the Volatility Bias

- Stable link  $y = \beta x + \varepsilon$  ( $\beta$  and  $\text{Var } \varepsilon$  fixed) with correlation  $\rho$ ; estimated only on days when  $\text{Var}(x)$ , the source market, is  $(1 + \delta)$  times higher, it gives (Forbes & Rigobon (2002))

$$\rho^* = \rho \sqrt{\frac{1 + \delta}{1 + \delta \rho^2}}, \quad |\rho^*| > |\rho| \text{ for } \delta > 0, \ 0 < |\rho| < 1$$

- Bitcoin – S&P 500, 2022–2026, days split by the VIX (top quartile:  $\text{VIX} > 21.8$ )
  - ▶ 0.52 on high-VIX days vs 0.32 on the others; S&P 500 variance ratio  $1 + \delta = 4.64$
  - ▶ adjusted  $\hat{\rho} = \rho^* / \sqrt{1 + \delta(1 - \rho^{*2})} = 0.27$
- The adjustment assumes no omitted common shock and no feedback between the two markets

## Appendix: Selection, Survivorship and Horizon Effects

- ▣ **Survivorship:** samples rebuilt from surviving firms and coins make past returns look too good (Brown, Goetzmann, Ibbotson & Ross (1992))
- ▣ **Selection:** Bitcoin's  $\times 257$  is chosen ex post among thousands of coins; a best-of-many Sharpe ratio needs a deflated SR
- ▣ **Horizon:** the depth of the maximum drawdown grows with the sample length
  - ▶ log price a driftless Brownian motion: the log drawdown  $D = \max_{s \leq t \leq T} (\ln P_s - \ln P_t) \geq 0$  has  $E[D] = \sqrt{\pi/2} \sigma \sqrt{T}$  (Magdon-Ismail et al. (2004))
  - ▶  $\sigma = 17.6\%$ :  $E[D] = 0.70$  over 10 years, 1.40 over 40; as price declines  $e^{-E[D]} - 1 = -50\%$  and  $-75\%$
  - ▶ the expected percentage MDD,  $E[e^{-D}] - 1$ , is smaller in magnitude (Jensen): about  $-47\%$  and  $-71\%$  in simulated paths
- ▣ The BET's  $-82.5\%$  (29 years) and an MDD from 11.7 years of data are not comparable without the same  $T$

## Appendix: ETF Premium Dynamics

- Statistical view of the arbitrage: an AR(1) premium

$$\pi_t = \phi \pi_{t-1} + u_t, \quad \text{half-life} = \frac{\ln 0.5}{\ln \phi}, \quad 0 < \phi < 1$$

- $\phi$  close to 1: premiums decay slowly, as for ETFs on foreign stocks or illiquid bonds ([Petajisto \(2017\)](#))
- In the US, in-kind redemptions also let the fund avoid distributing capital gains (a US tax rule, not a general feature)

## Appendix: Regulation of Digital Assets

|             | European Union                  | United States                                           |
|-------------|---------------------------------|---------------------------------------------------------|
| Framework   | MiCA, Regulation (EU) 2023/1114 | GENIUS Act, Public Law 119-27 for payment stablecoins   |
| Crypto ETPs | exchange-traded notes on crypto | spot Bitcoin ETPs approved in January 2024 (SEC (2024)) |
| Supervisors | ESMA, EBA, national authorities | SEC, CFTC, banking regulators                           |

- ESMA: European Securities and Markets Authority; EBA: European Banking Authority; SEC: Securities and Exchange Commission; CFTC: Commodity Futures Trading Commission
- BIS (Bank for International Settlements, the bank of central banks): BIS (2023) propose a “unified ledger” combining central bank money, deposits and assets; BIS (2025): stablecoins fall short of the three tests for money, **singleness** (accepted at par by all), **elasticity** (supply follows payment needs) and **integrity** (resists financial crime)

## Appendix: The Bucharest Stock Exchange, 1882–1948

- **1 Dec 1882:** the securities and commodities exchange opens under a royal decree of King Carol I [BVB \(history\)](#)
- Early shocks [BVB press release \(2017\)](#)
  - ▶ the 1907 peasants' revolt and the Balkan wars of 1912–1913
  - ▶ closed during the First World War
- After reopening: about 7 years of strong growth, then about 7 years of accelerated decline, overlapping the Great Depression of 1929–1933 [BVB press release \(2017\)](#)
- **1929:** a new law on exchanges, with the explanatory memorandum of minister Virgil Madgearu [Lege asupra burselor \(1929\)](#)
- **1948:** trading stops with the communist regime and the nationalisation of the economy [BVB \(history\)](#)



Share of UDR (Uzinele și Domeniile Reșița), June 1926. Scan: EDHAC e.V.; public domain;  
[Wikimedia Commons](#)

## Appendix: Palatul Bursei, the Exchange's Historic Home



Photo: Nicolae Ionescu (1928); public domain; Wikimedia Commons



Photo: Chainwit. (2023); CC BY 4.0; Wikimedia Commons

- Built 1906–1912, architect Ștefan Burcuș, Str. Ion Ghica 4 [Palatul Bursei](#)
- Seat of the Exchange and of the Bucharest Chamber of Commerce and Industry
- From 1955: the State Central Library (today the National Library)
- Left: stockbrokers at the entrance, March 1928; right: the building in 2023

## Appendix: Interruption and Rebirth, 1948–2023

- ▣ **1948–1995:** almost 50 years without a stock exchange [BVB press release \(2017\)](#)
- ▣ **20 Nov 1995:** first session of the re-established exchange: 905 shares of 6 listed companies [BVB press release \(2017\)](#)
- ▣ **1997:** launch of BET on 19 Sep 1997 at 1 000 points, with 10 companies; the basket may hold 10–20 and today has 20 [BVB \(indices\)](#)
  - ▶ **BET-FI:** the first sector index, for the financial investment companies (SIFs, listed in 1999)
- ▣ **2010:** BVB lists its own shares on its **regulated market** (an EU-authorized trading venue) [BVB \(history\)](#)
- ▣ **Sep 2020:** **FTSE Russell**, a global index provider, reclassifies Romania from Frontier to (Secondary) Emerging [BVB \(history\)](#)
- ▣ **Jul 2023:** Hidroelectrica lists through the largest IPO in BVB history, EUR 1.9 bn [BVB \(history\)](#)
- ▣ Largest daily BET move: 7 Jan 2009,  $-12.3\%$  ( $-13.1\%$  log), after  $+4.8\%$  the day before (official BVB daily archive [BVB \(indices\)](#))

## Appendix: Case Study, Data, Demand System and Estimation

- **Data:** quarterly Form 13F holdings, filed with the SEC by institutional managers with discretion over at least USD 100 million in 13(f) securities, 2001–2020; 80% of US market capitalisation (Section III.B)
  - ▶ characteristics  $X_k$ : log book equity, profitability, investment and dividend yield, each transformed to follow the Normal distribution, plus squared book equity; universe  $\mathcal{N}_i$ : stocks held in the past three years
- **Demand** (eq. 13–14),  $w_{ik}$  portfolio weight, 0 outside asset,  $p_k$  log market capitalisation:

$$\log(w_{ik}/w_{i0}) - p_k = \bar{d}_{0i} + \bar{d}'_{1i}X_k - \mathcal{E}_{ik} p_k + \xi \mathcal{E}_{agg,k} + \zeta' \mathcal{E}_{agg,k} X_k + \epsilon_{ik}$$

- ▶ active:  $\mathcal{E}_{ik} = \bar{\mathcal{E}}_{0i} + \bar{\mathcal{E}}'_{1i}X_k - \chi \mathcal{E}_{agg,k}$ ; passive:  $\mathcal{E}_{ik} = 0$
- ▶ equilibrium (eq. 17–18):  $\sum_i w_{ik} A_i = P_k$  and  $\sum_i (w_{ik} A_i / P_k) \mathcal{E}_{ik} = \mathcal{E}_{agg,k}$ ,  $A_i$  the investor's assets
- **Instruments:** price (eq. 23)  $\hat{p}_{k,i} = \log \sum_{j \neq i} A_j \mathbf{1}\{k \in \mathcal{N}_j\} / |\mathcal{N}_j|$ ; elasticity (eq. 24): the elasticity condition solved with the same equal-weighted portfolios
- **Estimator:** GMM with a nested fixed point solved by Newton's method (Section III.C); investors with fewer than 1 000 holdings pooled by AUM (assets under management) into groups of about 2 000 holdings (Table 2)

## Appendix: Case Study, Complete Results

| Topic              | Number in the paper                                                                                                                       | Where              | Reading                        |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------|--------------------|--------------------------------|
| Strategic response | $\chi = 2.97$ (SE 0.47); $\zeta = -0.48$ (SE 0.14)                                                                                        | Table 2, row 1     | between 0 and $\infty$         |
| Robustness         | $\chi$ from 2.30 to 3.27 in rows 2–14; 0.04 without the elasticity instrument, 0.63 without instruments                                   | Table 2            | endogeneity hides the response |
| Elasticity level   | mean $\mathcal{E}_{agg}$ 0.438 (fixed elasticities 0.39); residual cross-sectional SD 0.0395 (fixed 0.0842)                               | Table 3            | reactions halve the dispersion |
| Rise of passive    | median active share 81.4% (2001), 59.5% (2020)                                                                                            | Figure 5           | passive 19% to 41%             |
| Reduced form       | slope of $\Delta \log \mathcal{E}_{agg}$ on $\Delta \log$ active share 0.385 (SE 0.022); 0.336 (SE 0.055) with Russell 1000/2000 switches | Table 4, col. 2, 5 | consistent with one-third      |
| Prices             | per SD of elasticity, instrumented: volatility $-0.383$ (SE 0.123), illiquidity $-0.396$ (SE 0.131)                                       | Table 5, col. 2, 8 | inelastic: more volatile       |
| Evolution          | mean $\mathcal{E}_{agg}$ 0.41 to 0.27; with $\chi = 0$ the fall about twice as large                                                      | Figures 6, 8       | response damped the fall       |

## Appendix: Case Study, Replication with Course Data

- The 13F holdings are not in the course data, so  $\chi$  cannot be re-estimated; the arithmetic of Section V.A can
  - ▶ eq. (5) with  $\alpha = 70\%$  active,  $\chi = 3$ :  $0.70 + 0.30 \times 2.1/3.1 = 0.90$ ; eq. (28):  $1/(1 + 3 \times 0.68) = 0.33$
- Our measure (ETFs only), Financial Accounts of the United States: **ETFs' corporate equities over all sectors' corporate equities**
  - ▶ 0.4% (2001 Q1), 6.6% (2020 Q4), 10.3% (2026 Q2)
  - ▶ changes relative to 81%: active share  $-7.7\%$ , elasticity  $-2.5\%$  over 2001–2020;  $-12.3\%$  and  $-4.0\%$  to 2026 Q2
- The paper's broader measure adds passive mutual funds: 15 points over 30 years, i.e.  $-19\%$  of the 81% and  $-6.3\%$ ; the Financial Accounts do not split index and active mutual funds
- The authors also point to bank regulation, fire sales and large sovereign investors (Section VI)

## Appendix: A Minimal Python Workflow

```
import numpy as np, pandas as pd

URL = ('https://raw.githubusercontent.com/'
      'danpele/MFM/main/data/market/')

def load(sym, col='close'):
    d = pd.read_csv(URL + sym + '.csv',
                    index_col='date', parse_dates=True)
    return d[col].rename(sym)

spx = load('GSPC.INDX')           # index: close
spy = load('SPY.US', 'adjusted_close') # ETF
btc = load('BTC-USD.CC')          # 24/7
r_spx = np.log(spx).diff()        # one series
# joint analysis: join PRICES on common
# days first, then take returns
p = pd.concat([spx, btc], axis=1).dropna()
r = np.log(p).diff().dropna()
dd = spx / spx.cummax() - 1       # drawdown
```

- One series: returns on its own calendar, `np.log(s).diff()`
- Joint analysis: join the *prices* on common days, then take returns, so both cover the same interval (Friday → Monday)
- Returns taken before the join would pair a Bitcoin Sunday → Monday move with an S&P 500 Friday → Monday move
- Crypto trades every day: annualise with 365, not 252

## Appendix: Data Checklist

- Plot the raw series and list the largest returns
- Compare two sources when possible
- Check adjustments: dividends, splits, redenominations
- Remove spurious non-trading-day records; investigate isolated outliers, and correct or remove only confirmed data errors; document every change
- For joint analyses, join the prices on common dates before computing returns; for a single asset, use its own calendar
- Watch survivorship: delisted firms vanish from today's index lists

## Appendix: The Discovery Loop, Reusable Prompts (1/2)

### ▣ Literature

- ▶ List peer-reviewed papers since 2018 on passive ownership and price informativeness. For each: DOI, identification strategy, main estimate. Flag any paper you are not sure exists.
- ▶ Resolve every DOI on Crossref; title and authors must match; read the abstract yourself

### ▣ Hypothesis

- ▶ Give three competing explanations for SPY beating RSP after 2015 (passive flows, sector composition, earnings growth of the largest firms). For each, one testable implication the others do not share.

### ▣ Data and code

- ▶ Write Python that reads the daily adjusted (total-return) prices of SPY and RSP from the course data, joins prices on common days, and tests  $H_0$ : mean daily log-return spread = 0 with a Newey-West t-statistic.

## Appendix: The Discovery Loop, Reusable Prompts (2/2)

### ▣ **Robustness**

- ▶ Write a pre-registered robustness plan: subsamples, HAC bandwidths, simple vs log returns, a sector-neutral version. State in advance which outcome would falsify the hypothesis.

### ▣ **Critique**

- ▶ Act as a hostile referee. List the three strongest threats to identification in this design and the test that addresses each.

### ▣ **Report:** estimate, standard error, sample, every specification tried, also the failed ones

- ▣ What the researcher must check: invented references (resolve each DOI); unadjusted instead of adjusted close (reproduce the total-return multiples first); returns before the join (count the common days); window shopping (fix the windows in advance); low power (report the MDE)

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